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13 December 1984
SUPERSEDING
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MILITARY SPECIFICATION

TRUCK CONCRETE, MOBILE MIXER,

8 CUBIC YARD,

COMMERCIAL CONSTRUCTION EQUIPMENT (CCE)

This specification is approved for use by the USA Belvoir Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a truck-mounted, 8 cubic yard, mobile concrete mixer.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and, standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-B-601	- Boxes, Wood, Cleated-Plywood.
PPP-P-40	- Packaging and Packing of Hand Tools.

MILITARY

MIL-P-116	- Preservation, Methods of.
MIL-E-10062	- Engines: Preparation for Shipment and Storage of.
MIL-G-20241	- Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.

Beneficial comments (recommendations, additional, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 3895

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MIL-P-23377	- Primer Coating, Epoxy Polyamide, Chemical and Solvent Resistant.
MIL-E-52798	- Enamel, Alkyd, Camouflage.
MIL-E-52835	- Enamel, Modified Alkyd, Camouflage, Lusterless.
MIL-S-81733	- Sealant And Coating Compound, Corrosion.

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-209	- Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment.
MIL-STD-281	- Automobile, Trucks, Truck-Tractors and Trailer Dollies, Preservation and Packaging of.
MIL-STD-642	- Identification Marking of Combat and Tactical Transport Vehicles.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1474	- Noise Limit for Army Materials.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

MIL-T-PD-9156	- Truck Chassis, Truck Tractors and Trucks: Commercial, Diesel Engine Driven, 50,000 to 5,000 Pounds GVWR, 6 x 4, 6 x 6, and 8 x 6; XM915, XM916, XM917, XM918, XM919 and XM920.

DEPARTMENT OF LABOR (DOL)

Occupational Safety and Health Administration (OSHA) Standards.

29CFR 1910	- Occupational Safety and Health Administration (OSHA) Standards For Construction Jobs.
29CFR 1926	- Construction Safety Standards.

(The code of Federal Regulations [CFR] and the Federal Register [FR] are for sale on a subscription basis by the Superintendent of Documents, U.S. Government printing Office, Washington, DC 20402. When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

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(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASSOCIATION OF AMERICAN RAILROADS (AAR)

Rules Governing the Loading of Commodities on Open Top Cars of the Association of American Railroads.

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 commonwealth Drive, Warrendale, PA 15096.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. This specification covers a truck-mounted, mobile concrete mixing plant capable of transporting unproportioned and unmixed materials for producing 8 cubic yards of mixed concrete. The mixer shall be mounted on a commercial truck chassis as specified herein (see 5.4). The mixer shall be complete with all components furnished as standard equipment that are customarily offered with a mixer of this type whether stipulated herein or not, together with such attachments as may be necessary to enable the mixer to function reliably and efficiently in a sustained operation. Established component ratings shall not be increased to meet the requirements of this specification. Overall width of the mixer body shall not exceed 98 inches.

3.1.1 Environmental conditions. The mixer truck shall function and travel as specified herein under worldwide climatic conditions from 25^0 F to $+125^0$ F and from sea level to a 5000 foot altitude. The mixing plant shall function between $+32^0$ F through $+125^0$ F.

3.2. First article. Unless otherwise specified (see 6.2), the sample shall be subjected to first article inspection (see 4.3.1 and 6.3). Any change or deviations of production truck-mounted mixers from the approved first article during production will be subject to the approval of this

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contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish truck-mounted mixers conforming to this specification.

3.3 Mixer construction.

3.3.1 General. The mixer shall meet the following physical characteristics:

Minimum materials carrying capacity	- Cement	- 63 cubic feet
	- Sand	- 128 cubic feet
	- Water	- 400 gallons
	- Aggregate	- 182 cubic feet

Each material shall be carried in a separate compartment.

3.3.2 Performance. The mixer shall be capable of continuously proportioning and mixing quality concrete at a rate of 1 cubic yard per 2-1/2 to 4 minutes, depending on mix design.

3.3.3 Bins and tanks. The bins and tanks shall have the capacities specified in 3.3.1. The sand and aggregate bins shall have open tops and shall be capable of being charged by front end loaders or overhead bins. The sand and cement bins shall each be equipped with a pneumatic vibrator to assure continuous flow of materials. The cement bins shall be of the close-top type. A tarpaulin of sufficient size to cover the sand and aggregate bins shall be provided. Tiedown points of the bin and the tarpaulin shall be provided. The sand and aggregate bins shall be covered with screens to keep oversize rocks from entering the bins and mixing area.

3.3.4 Quick loading cement hopper. A hopper shall be provided to fit over the standard opening in the top of the cement bin. The hopper shall be provided with a vibrating screen and hinged cover. The cover shall cover the entire hopper opening. The hinged cover shall be provided with a circular opening to accept an elephant-trunk from an overhead bin. The opening shall be provided with a hand operable, removable cover.

3.3.5 Water system. Water shall be supplied by a 400 U.S. gallon mounted tank(s). The tank(s) shall be coated on the inside with corrosion-resistant material in accordance with the manufacturer's standard practice. A water fill cap secured to the tank(s) by a chain shall be provided for each fill opening. The inside diameter of the fill port shall not be less than 2 inches and the fill port shall be labeled "Water". A water level gage of the sight tube type, graduated in U.S. gallons, shall be provided and this gauge shall be mounted or protected in a manner which prevents breakage. A water pump shall be provided with the water system. A washout hose shall be provided to clean the mixing auger and other critical areas. A water control valve calibrated in U.S. gallons shall be provided to control the amount of water being added to the mix.

3.3.6 Mixing system. Mixing of the concrete shall be by an auger in a reinforced rubber chute. The auger shall be equipped with replaceable wear plates. The mixing auger shall be powered. A system shall be provided to raise and lower the mixing chute. The mixing chute shall

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be equipped with an anti-splash cover to facilitate clean-out and provide clean operation. Two delivery chutes shall be provided with supports and hand operable fasteners to attach to the mixing chute.

3.3.7 Mix proportioning system. The continuous proportioning of materials shall be controlled by volume. Adjustable volume control gates shall be furnished on the sand and aggregate bins for proportioning the mix to meet desired design strengths. The cement shall be proportioned by a rotary feeder encased in a drop-through housing. The sand and aggregate shall be conveyed to the mixing chamber from the respective bins by a common conveyor belt. The conveyor belt shall be moved by two paralleled chains adequately supported by removable wear bars. A system to continuously oil the chains during mixing shall be provided. If the chain oiling system must be attended on a less than weekly basis, then the oiling system must be readily accessable.

3.3.8 Mixer drive train. Power for the mixer shall be provided from the truck engine or transmission through a power take-off. Power take-off speed shall be proportional-to-engine speed. The power take-off shall permit engagement without clashing with the engine running and transmission in neutral. The power from the power take-off shall be transmitted to power hydraulic pump and water pump. Power take-off and controls are provided by the mobile mixer manufacturer.

3.3.9 Quick-load cement bin. A portable bin shall be provided into which bags of cement may be emptied. The bin shall be so designed that it may be lifted by a front end loader and the cement in it may be dumped into the cement bin on the mixer. The quick-load cement bin shall have a capacity of at least 16 cubic feet.

3.3.10 Admixture system. The manufacturer shall supply his commercial liquid and dry cement admixture systems.

3.3.11 Air system. The system provided to supply air pressure for the pneumatic vibrators shall be taken from the air brake system of the truck chassis. Assure that low pressure release of truck air brakes will not occur.

3.4 Truck chassis. The mixer shall be mounted on a commercial truck chassis as specified in MIL-T-PD-9156.

3.5 Lubrication. The truck-mounted mixer shall be capable of operating using standard military lubricants. All components requiring lubrication shall be provided with means of lubrication. Initial servicing of each truck-mounted mixer shall be with military lubricants applicable to this type of equipment.

3.6 Maintenance.

3.6.1 Maintainability. All major assemblies and installed attachments shall be accessible for maintenance, repair, and replacement with minimum removal of other major assemblies and

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installed attachments. Covers or plates which must be removed for component adjustment or maintenance shall be equipped with quick-disconnect fastenings. Drain outlets shall be accessible. Ease of maintenance provisions shall incorporate features insuring operating clearances for handtools and lubricating equipment, and the accessibility for service operations such as adding oil to the engine crankcase and adding oil to gear cases. Each maintenance, assembly, or disassembly operation required during testing in accordance with 4.3 shall be accomplished by no more than two men (except a hoist and operator may be used as necessary) using general purpose tools normally associated with equipment of this nature. After completion of any maintenance operation, the truck-mounted mixer shall operate as specified herein.

3.6.2 Interchangeability. All parts having the same part number shall be functionally and dimensionally interchangeable. Common hardware and fastening devices such as bolts, nuts, screws, and washers shall be held to a minimum number of types and sizes. All screw threads shall be in accordance with FED-STD-H28.

3.7 Safety. All reciprocating, rotating or moving parts of equipment shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard. All hot surfaces of equipment, including exhaust pipes and other lines which may be subject to high operating temperatures, exposed to contact by personnel or which create a fire hazard shall be fully guarded or insulated. Exhaust or discharges from equipment shall be directed so that they do not endanger personnel or obstruct the view of the operator. Electrical equipment shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazard. Fuel tanks shall be located in a manner which will not allow spills and overflows to run onto the engine exhaust, burners, or electrical equipment. Nonfunctional sharp edges, projecting points, and excessive length of fastening devices shall be avoided. Steel walkways and ladders shall be provided where necessary for inspection, servicing, and maintenance. Walkways and ladders shall provide safe and unobstructed access and egress. All platforms, steps, and ladder rungs shall have anti-skid surfaces. The truck-mounted mixer shall meet the applicable requirements of the Occupational Safety and Health Administration standards. (29CFR 1910 and 29CFR 1926). A type B back-up alarm conforming to the performance criteria of SAE J994 shall be provided.

3.8 Noise. Truck-mounted mixer shall not exceed MIL STD-1474 (see 5.1.1.1) category D [85db(A)], at the operators' position or other occasionally occupied positions, when tested in accordance with 4.3.2.14. If technical infeasibility is established via MIL-STD-1474 (see 5.1.1.2), or the means of reduction is too costly, the written permission of the procuring activity shall be requested to exceed the 85db(A) limit. The precautions of MIL-STD-1474 shall be provided if and only if MIL-STD-1474 (see 5.1.1.2) procedures have been pursued and documented to the satisfaction of the procuring activity (see 6.5). Hazard signs shall conform with MIL-STD-1474 (see 4.3) and be readable at the distance indicated on the sign.

3.9 Material deterioration and control. The truck-mounted mixer shall be fabricated from compatible materials, inherently corrosion resistant or treated to be resistant to the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environment to which the item may be exposed.

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3.9.1 Dissimilar metals. Dissimilar metals as defined in MIL-STD-889 shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation may be provided by a barrier such as a coat of epoxy primer conforming to MIL-P-23377, or corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241.

3.9.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components or sub-components, and shall make information available, upon request, to the contracting officer or designated representative.

3.9.3 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the truck-mounted mixer may be newly fabricated from recovered materials to the maximum extent practicable, provided the truck-mounted mixer produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the truck-mounted mixer

3.10 Slinging attachments. Permanently affixed slinging or multipurpose (class 1 or class 3) attachments that enable the truck-mounted mixer to be lifted in its normal travel position shall be provided and labeled ”Lift Here”. Each attachment shall meet the requirements specified in MIL-STD-209 for type II equipment.

3.11 Equipment tiedown attachments. Permanently affixed tiedown or multipurpose attachments that permit fastening the truck-mounted mixer to the floor or deck of a transportation medium shall be provided, and labeled "Tiedown Here". The tiedown attachments (class 2 or class 3) shall meet the requirements specified in MIL-STD-209 for type II equipment.

3.12 Treatment and painting. The parts of the truck-mounted mixer normally painted shall be cleaned, treated, and painted in accordance with the manufacturer's standard commercial practice. The final color shall be lusterless forest green in accordance with MIL-E-52798 or MIL-E-52835.

3.13 Marking. Marking and registration shall be in accordance with MIL-STD-642, color lusterless black.

3.14 Identification and instruction plates. The manufacturer shall provide all identification and instruction plates normally furnished. The plates shall be permanently affixed to the truck-mounted mixer in a conspicuous location.

3.15 Transportability. The complete vehicle shall be transportable with operational equipment installed.

3.15.1 Aircraft cargo. The vehicle shall be transportable in a C-5A aircraft.

3.15.2 Railroad loading. The truck-mounted mixer shall be rail transportable on a flatcar to comply with the requirements of the Rules Governing the Loading of Commodities on Open Top Cars of the Association of American Railroads.

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3.16 Human factors engineering. The following Society of Automotive Engineers specifications shall apply: J1500 (Universal Symbols), J614 (Dipstick Marking), J925 (Access Dimensions), J209 (Instrument Face Design). The mixer shall be designed to be operable and maintainable by 5th through 95 percentile personnel (see SAE J833) who are dressed appropriate to the environments of interest (32° F to 125°F). All valves shall be labeled open/close, arrows shall be used to indicate direction of rotation, and all valves shall be labeled with function served by the valve, i.e., "water valve" rather than valve 3. All operator relevant components shall be labeled in terms that communicate to the operator.

3.17 Workmanship.

3.17.1 Steel fabrication. The steel used in fabrication shall be free from kinks, sharp bends, and other conditions which would be deleterious to the finished product. Manufacturing processes shall not reduce the strength of the steel to a value less than intended by the design. Manufacturing processes shall be done neatly and accurately. All bends shall be made by controlled means to insure uniformity of size and shape.

3.17.2 Bolted connections. All bolt holes shall be accurately punched or drilled and shall have the burrs removed. Washers or lock-washers shall be provided. All nuts, bolts, and screws shall be tight.

3.17.3 Riveted connections. All rivet holes shall be accurately punched or drilled and shall have the burrs removed. Rivets shall completely fill the hole. Rivet heads shall be of an approved shape and shall be concentric with the rivet. Rivet heads shall make full contact with the intended surface. The rivets shall be so fabricated as to develop a joint strength not less than the design value.

3.17.4 Welding. Surfaces of welds shall be free from imperfections which will be injurious to the weldments. Welding procedures shall be in accordance with a nationally recognized code. The contractor shall make available to the contracting officer certification that each manual welder and automatic welding equipment operator has a current qualification as proscribed by either AWS or ASME code. Welds shall be of sufficient size and shape to develop the full design strength of the parts connected by the welds. Welds shall transmit imposed stresses without permanent deformation or failure when subjected to proof or service loadings.

3.17.5 Listings. All castings shall be sound and free from patching, misplaced coring warping, or any other defects which reduces the castings ability to perform its intended function.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection comparison (see 4.5).
d. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 First article inspection. Unless otherwise specified in the contract (see 6.2), the preproduction truck-mounted mixer shall be examined as specified in 4.4 and tested as specified in 4.3.2.1 through 4.3.2.14 to determine conformance to the requirements of this specification. Presence of one or more defects shall because for rejection. Failure of any test shall be cause for rejection.

4.3.2 Tests.

4.3.2.1 Test conditions. Unless otherwise specified in a test, tests shall be performed without shelter and at the climatic conditions existing at the place of test. The truck-mounted mixer shall operate as specified herein without maintenance other than the contractor a recommended normal scheduled maintenance as established by a maintenance schedule prepared and submitted by the contractor prior to test.

4.3.2.2 Wading. Demonstrate the ability of the mixer bins to be loaded by a front end loader by filling the sand and aggregate bins with sand and aggregate respectively with a front end loader. Inability of the mixer bins to be loaded by the front end loader shall constitute failure of this test.

4.3.2.3 Mounting. With all bins and the water tank loaded to capacity, test the mounting of the mixer on the chassis by accelerating from stop to 10 miles per hour as quickly as possible, then applying brakes to stop suddenly. Repeat this 10 times. Accelerate backward from rest to maximum reverse speed as quickly as possible, then apply the brakes to stop suddenly: repeat this 10 times. Evidence of permanently deformed or broken mounting devices or misalignment between chassis and mixer shall constitute failure of this test.

4.3.2.4 Mixing ability. Demonstrate the mixing ability of the mixer by setting the controls for a six bag mix and producing a minimum of one cubic yard of concrete. Determine the time required to produce one cubic yard of concrete. Examine the fresh concrete to determine if it is a uniform mix. Examine the aggregate for complete coating with cement paste. Inability to produce a uniform mix or to completely coat the aggregate shall constitute failure of this test.

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Inability to produce one cubic yard of six bag mix concrete within four minutes shall constitute failure of this test. Demonstrate the ability of the mixer to produce various design strength mixes by setting the controls for a four bag mix and producing a minimum of one cubic yard of concrete. Examine the fresh concrete to determine if it is uniform mix. Examine the aggregate for complete coating with cement paste. Inability of the mixer to produce a uniform mix or to completely coat the aggregate shall constitute failure of this test.

4.3.2.5 Axle loading. Determine, by weighing, the load on each individual axle under the following conditions:

a. The water supply tank filled to capacity (see 3.3.1).
b. The coarse aggregate bin filled to capacity (see 3.3.1). The weight of the aggregate shall be not less than 100 pounds/cubic foot.
c. The sand bin filled to capacity (see 3.3.1). The weight of the sand shall be not less than 95 pounds/cubic foot.
d. Cement bin filled to capacity (see 3.3.1).
e. Wet and dry admixture systems filled to capacity.
f. All reservoirs, to include fuel tank(s), filled to capacity.

4.3.2.6 Highway speed. Demonstrate the truck's maximum high-gear, highway speed capability on hard surfaced roadway with the vehicle loaded as specified in 4.3.2.5. Average the speed for four runs, two in each direction.

4.3.2.7 Gradeability. Demonstrate the gradeability of the truck mixer when loaded as specified in 4.3.2.5. Demonstrate driving the truck mixer up no less than a 25 percent grade, stop and park the truck, restart and continue ascending the slope. Demonstrate the ability of the truck mixer when loaded as specified in 4.3.2.5 to ascend a 3 percent grade at no less than 20 miles per hour. Demonstrations shall be in accordance with SAE J950.

4.3.2.8 Mixer truck mobility. Mixer truck mobility shall be performed as specified herein. Permanent deformation, overheating, malfunction, failure to complete any test, or leakage of lubricants or hydraulic fluids shall be cause for rejection of the first article.

Terrain	Speed	Loading
Primary highway types	50 miles at up to max mph	As specified in 4.3.2.5
Primary highway types	50 miles at up to max mph	Empty
Secondary roads	25 miles at up to 60 percent max. mph	As specified in 4.3.2.5
Secondary roads	25 miles at up to 60 percent max. mph	Empty
Off-road terrain	10 miles at safe speeds	As specified in 4.3.2.5
Off-road terrain	5 miles at safe speeds	Empty

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During the operation, demonstrate the following:

a. Ability to load, transport, and unload.
b. Based on the primary highway fuel consumption both loaded and empty, calculate travel distance per tank full.
c. Rocking of the mixer truck with load in soft, slippery, off-road terrain by cycle shifting at not more than 2-second intervals from forward to reverse and back, continuously for at least 60 seconds. Achieve at least 3/4 wide-open-throttle power in each gear. Repeat the load.

4.3.2.9 Turning capability. Demonstrate vehicle clearance circle and curb clearance circle in accordance with SAE J695, Field Test Procedures.

4.3.2.10 Braking-capability. Demonstrate the brake capability of the mixer truck. Conduct service, emergency, and parking ability demonstrations.

4.3.2.11 High temperature operation. The contractor shall furnish test data to prove that the truck-mounted mixer is capable of operating in an ambient temperature of 120°F without exceeding the engine manufacture's recommended limits for radiator top tank temperature, engine sump and gallery oil temperatures, and hydraulic component manufacturer's hydraulic fluid temperature limits.

4.3.2.12 Test procedures of slinging provisions. Test procedures of slinging provisions shall be as follows:

a. Lift the truck-mounted mixer and hold in suspension in its normal travel position using slings that converge not more than 24 feet above the lowest extremity of the truck-mounted mixer. Determine the true angle of the sling legs from the vertical. Measure the clearance between each sling and the truck-mounted mixer. Measure eye openings and clearance dimensions of the provisions. Measure the static load carried by each provision.
b. Restrain the truck-mounted mixer by anchoring the main frame and subject each provision to a working load force equal to the load factor specified in MIL-STD-209 times the load it carried in "a" above. An alternate method of test is to lift the truck-mounted mixer as in "a" above and add weights to the main frame until the required force is obtained. Hold the truck-mounted mixer under either of the above loading conditions for at least 90 seconds.

4.3.2.12.1 Failure criteria. Weld failure or permanent deformation caused by lifting or failure to meet the requirements of MIL-STD-209 shall constitute failure of this test.

4.3.2.13 Test procedures of equipment tiedown provisions. With the truck-mounted mixer anchored by means other than the provisions being tested, apply the distributed loads as specified in MIL-STD-209 to each attachment. Measure dimensions of the tiedown.

4.3.2.13.1 Failure criteria. Failure to meet the requirement of MIL-STD-209 shall constitute failure of this test.

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4.3.2.14 Noise level testing requirements. The equipment noise level at operator's station and all immediate occupied areas shall not exceed 85dB(A) when tested in accordance with the requirements of 5.1 of MIL-STD-1474. The data shall include dB(A) levels taken at the operator's station and all occasionally occupied positions. Exterior noise levels for drive-by shall not exceed 88dB(A) when tested in accordance with the requirements of 5.1 and 5.2 of SAE J1077. If the data exceeds the limits specified in category D, table 2 of MIL-STD-1474, for equipment noise level, or 88dB(A) for drive-by, the contractor shall reduce the noise level or provide documentation verifying that compliance is either technically or economically unfeasible.

4.3.2.14.1 Failure criteria. Failure to reduce the noise level or provide documentation verifying that compliance is either technically or economically unfeasible shall constitute failure of this test.

4.4 Quality conformance inspection.

4.4.1 Examination. Each truck-mounted mixer shall be examined for the following characteristics. Presence of one or more defects shall be cause for rejection.

101.	Mixer construction not as specified.	3.3
102.	Hopper not as specified.	3.3.4
103.	Bins and tanks not as specified.	3.3.3
104.	Water system not as specified.	3.3.5
105.	Mixing system not as specified.	3.3.6
106.	Mixer drive train not as specified.	3.3.8
107.	Quick loading cement bin not as specified.	3.3.9
108.	Admixture system not as specified.	3.3.10
109.	Air system not as specified.	3.3.11
110.	Truck chassis not as specified.	3.4
111.	Lubrication not as specified.	3.5
112.	Ease of maintenance not as specified.	3.6
113.	Maintainability not as specified.	3.6.1
114.	Interchangeability not as specified.	3.6.2
115.	Safety not as specified.	3.7
116.	Materials are not resistant to corrosion or deterioration or treated to be made resistant to corrosion or deterioration for the applicable storage and operating environment.	3.9
117.	Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other	3.9.1
118.	Contractor does not have documentation available for identification of material, material finishes or treatments.	3.9.2
119.	Sling attachment not as specified.	3.10
120.	Tiedown attachments not as specified.	3.11
121.	Treatment and painting not as specified.	3.12
122.	Marking not as specified.	3.13

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123.	Identification and instruction plates not as specified.	3.14
124.	Transportability not as specified.	3.15
125.	Aircraft cargo transportability not as specified.	3.15.1
126.	Railroad transportability not as specified.	3.15.2
127.	Human factors not as specified.	3.16
128.	Workmanship not as specified.	3.17
129.	Steel fabrication not as specified.	3.17.1
130.	Bolted connections not as specified.	3.17.2
131.	Riveted connections not as specified.	3.17.3
132.	Welding not as specified.	3.17.4
133.	Casting not as specified.	3.17.5

4.4.2 Operational run-in. Each item of equipment shall be serviced as required and shall be run-in in accordance with the manufacturer's normal procedures for a period of not less than 60 minutes. One or more functional defects shall constitute failure and acceptance of the equipment shall not be approved until all defects are corrected.

4.5 Inspection comparison. The Government may select truck-mounted mixers at any time during the contract production period and subject these truck-mounted mixers to the examination specified in 4.4.1 and the tests specified in 4.3.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government, on units selected at random from those which have been accepted by the Government and will not include the previously inspected preproduction truck-mounted mixer and initial production truck-mounted mixer. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.5.1 Inspection failure. Failure of an inspection comparison truck-mounted mixer to meet any requirement specified herein during and as a result of the examination and tests specified in 4.3.2 shall be cause for refusal by the Government to continue acceptance of production truck-mounted mixers until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on truck-mounted mixers previously accepted and produced under the contract. Any deficiencies found as a result of the insection comparison will be considered prima facie evidence that all truck-mounted mixers accepted prior to completion of inspection comparison are similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.6 Inspection of packaging.

4.6.1 Quality conformance inspection.

4.6.2 Unit of product. For inspection purposes a complete concrete mixer prepared for shipment shall be the unit of product.

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4.6.3 Examination. Each mixer shall be examined for the following defects. Any areas of noncompliance with the packaging requirements of this specification shall be corrected before the mixer is shipped.

134. Truck chassis not preserved as specified for level A.
135. Engine not preserved as specified for levels A and C.
136. Functional components with critical surfaces not preserved for level A.
137. Bins and tanks not dry and covered with tarpaulin as specified for level A.
138. Washout hose not coiled, secured, and placed in storage space or boxed as specified for levels A and C.
139. Hydraulic system not filled as specified.
140. Conveyor belts and V-belts not loosened if required for level A.
141. Air system not purged of water as specified.
142. Special tools not preserved and placed in tool box as specified.
143. Removed components not packed and secured to mixer as specified for level A.
144. Marking missing, improper, or illegible for levels A and C.
145. Depreservation guide not provided as specified

5. PACKAGING

5.1 Preservation. Preservation shall be level A or C as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Disassembly. Any components which increase the shipping cube of the mixer and which may be easily removed and replaced, shall be removed. Such components shall be boxed if necessary to prevent damage and pilferage, and shall be secured to the mixer in a location which does not increase the cube. If such components cannot be secured on the mixer, they shall not be removed.

5.1.1.2 Truck chassis. The truck chassis shall be preserved in accordance with the level A requirements of MIL-STD-281 for mobile shipment.

5.1.1.3 Engine. The engine and related accessories shall be preserved in accordance with MIL-E-10062, type I, method I, level A. The engine cooling system shall be preserved using the water and antifreeze procedure.

5.1.1.4 Mixer. Any components of the mixer which are not corrosion resistant and which are critical to the proper operation of the mixer shall be preserved. Preservation shall be in accordance with the guidance contained in MIL-P-116 and shall utilize the "P" designated preservatives of that specification.

5.1.1.4.1 Bins and tanks. Bins and tanks shall be clean and dry. The tarpaulin cover provided shall be put in place over the bins and secured as intended.

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5.1.1.4.2 Water system. The tank and other parts of the water system shall be drained. The washout hose shall be coiled and secured with suitable ties.

5.1.1.4.3 Hydraulic system. The hydraulic system shall be filled to the operating level with the designated operating hydraulic fluid. The system shall be operated to insure coverage of all internal surfaces.

5.1.1.4.4 Conveyor belts and V-belts. Conveyor belts and V-belts shall have their tension relaxed unless the manufacturer recommends otherwise. A tag shall be affixed near each conveyor or belt informing that the items are loose and must be tightened to the specified degree prior to operation of the mixer. The chains which move the conveyor belt shall be preserved with P-11 as listed in MIL-P-116.

5.1.1.4.5 Air system. The air system shall be completely drained or purged of any water and the drains left open if applicable.

5.1.1.4.6 Quick load cement bin. The portable, quick load cement bin shall be preserved in accordance with MIL-P-116.

5.1.1.4.7 Special tools. If provided, special operating or maintenance tools shall be preserved in accordance with PPP-P-40 or with the criteria outlined in MIL-P-116.

5.1.2 Level C. Each complete mixer shall be preserved in a manner which will protect critical, functioning components and accessories from corrosion during the interval between the production line and the first destination. The engine cooling system shall be preserved in accordance with 5.1.1.3. The engine crankcase and all other fluid reservoirs shall be filled to the operating level with the approved lubricants and fluids. The tarpaulin shall be in place over the bins and tanks.

5.2 Packing. Packing shall be level A or C as specified (see 6.2).

5.2.1 Level A. The mixer shall be shipped mobile and require no additional packing except for the securement to the mixer of any removed and boxed components such as the washout hose, portable cement bin, and delivery chutes and their supports. Such items shall be secured in compartments provided, if such exist. Boxes shall comply with PPP-B-601, overseas type. Special tools shall be placed in the tool box and the box secured to prevent accidental opening.

5.2.2 Level C. The mixer shall require no additional packing for mobile shipment.

5.3 Marking. Marking for shipment and storage shall be in accordance with MIL-STD-129.

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5.4 Depreservation guide. A depreservation guide shall be prepared and placed in a waterproof envelope. The guide shall be secured to the steering wheel of the mixer. Unless otherwise specified (see 6.2), DA Form 2258, Depreservation Guide for Vehicles and Equipment shall be used (see 6.6).

6. NOTES

6.1 Intended use. The truck-mounted mixer is intended for use in mixing concrete.

6.2 Ordering data. Procurement documents should specify the following:

a. Title, number and date of specification.
b. Date of issue of DoDISS applicable and exceptions thereto (see 2.1.1).
c. When first article inspection is not required (see 3.2).
d. Time frame for submission of first article truck-mounted mixer and number of truck-mounted mixers required (see 3.2).
e. When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests shall be conducted by the contractor (see 3.2).
f. When first article truck-mounted mixer inspection is not required (see 3.2).
g. Level of preservation and of packing required (see 5.1 and 5.2).
h. When other than DA Form 2258 is to be used (see 5.4).

6.3 First article. When a first article inspection is required, the items shall be a production model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examination, tests, and approval of the first article test results and dispositions of the documents first article.

6.4 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools as necessary (including any special tools), and instructions on shipment of the truck-mounted mixers.

6.5 Noise limits. When the noise level of the truck-mounted mixer exceeds 85 dB(A), the appropriate discussion of the noise hazard in accordance with MIL-STD-1474 (MANUALS paragraph) shall include the requirement for hearing protection, the noise level of the truck-mounted mixer and the distance from the truck-mounted mixer at which 85 dB(A) will always be met.

6.6 Depreservation guides. The contracting officer should arrange to furnish the necessary copies of DA Form 22589 when requested by the contractor (see 5.4).

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